

Task 8 — Courses of Action

Constrained trajectories on the admissible submanifold

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Track	Algorithm-only, unclassified, laptop-based evaluation
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Version	FormSpace overlay v1.0 — 2026-09-01

1. Overview — COAs as Constrained Trajectories in Form Space

A Course of Action is a proposed trajectory through Form Space, filtered by the deterministic admissibility gate $G(E, \varphi, v, A, R) \rightarrow \{\text{PASS, REVIEW, DENY, PAUSE}\}$, where E is evidence sufficiency, φ is policy, v is environmental stewardship, A is the action envelope, and R is the rollback plan. MYCA proposes; AVANI gates; human disposition finalizes. No COA passes the gate if it exits the admissible submanifold or lacks a defined rollback.

2. The Admissibility Gate

The gate is a deterministic function, not a learned classifier. Its inputs are explicit records:

- Evidence sufficiency E : independent-source count, modality count, calibration recency, uncertainty envelope.
- Policy φ : allowed action types for the current mission, authority level, operator role.
- Environmental stewardship v : exclusion zones, geofences, environmental limits, permit constraints.
- Action envelope $A = (\text{identity, scope, magnitude, duration, preconditions, stop conditions, rollback, authority})$ — the operator-visible field on every COA.
- Rollback plan R : reversible actions preferred; irreversible actions require higher authority and a written stop condition.

3. MYCA Seven-Agent Loop

- State / evidence summarizer.
- COA proposer.
- Operations critic.
- Safety / AVANI critic.
- Communications / energy critic.
- Evidence verifier.
- Final synthesizer.

Protocol: structured state $\rightarrow$ independent proposals $\rightarrow$ evidence check $\rightarrow$ critiques $\rightarrow$ revision (bounded to max 2 iterations) $\rightarrow$ scoring $\rightarrow$ AVANI gate $\rightarrow$ human disposition. Scoring uses a log-opinion pool with lineage-aware weights per the NLM scientific architecture.

4. Comparison Runs

- Single-agent COA generation (baseline).
- Multi-agent generation without evidence verifier.
- Full MYCA process (all seven agents).
- Full process with one modality removed.

Headline metric distinguishing full MYCA from single-agent: unsupported-assertion rate — the fraction of COA justifications that do not trace to a cited evidence ID in MINDEX.

5. Outputs

- Ranked COA set with per-COA action envelope.
- Explicit rollback plan and stop conditions.
- Evidence-cited justifications (every claim → MINDEX ID).
- AVANI gate disposition per COA (PASS / REVIEW / DENY / PAUSE).
- Human-authorization capture for any executed COA.

6. Metrics

- Time to converge (bounded at ≤ 240 s live).
- Completeness (fraction of required envelope fields populated).
- Unsupported-assertion rate.
- Safety-policy compliance rate.
- Human SME rating on a 5-point rubric.

7. Timing Budget (live, on-site)

- Cold start ≤ 30 s.
- Full COA convergence ≤ 240 s.
- Integrated Task 12 + 13 + 8 end-to-end ≤ 420 s.

8. Governance Non-Claims

- No autonomous high-risk actuation. Every high-risk action requires human disposition.
- No override of local device safety controllers or flight controllers.
- No claim of general reasoning — MYCA operates on the structured state + evidence graph produced by Tasks 12 and 13.

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